

Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 23 papers, 2069–2082 BS

2. Data Preprocessing

6 hrs · in 23/23 papers · 4 topics, 2 TOP tier · **295 marks, 15.2% of the archive** · **49 numerical (17%) + 246 theory** · first chapter with numericals
→ see the Ch 2 Numerical companion.

Weight of the chapter:

- Data Pre-processing **TOP 12/23** → most-asked topic in the whole syllabus.
- OLAP & Multidim Analysis **TOP 10/23**
- Similarity Measures **HOT 6/23** → nearly always a numerical.
- Data Types & Attributes **PIN 1/23**

2.1 Data Types and Attributes

What is DM? Explain diff data types of attributes in a dataset. **10**

PIN 1/23 71 Shr DM definition → Ch 1. Only attribute types needed here.

Attribute

- Property / characteristic of an object. Eg eye colour, temperature.
- Also called: variable, field, feature, characteristic.
- **Object** = record, point, case, sample, entity, instance.
- Same attr → diff value sets (height in ft or m).
- Diff attrs → same value set, diff properties (ID and age are both int, but ID has no max).

Han & Kamber 3rd ed: 4 types (use this, it is the syllabus ref book)

- **Nominal**: names / categories only, no order. Zip code, employee ID, eye colour.
- **Binary**: nominal w/ exactly 2 states (0/1).
 - **Symmetric**: both states equally valuable. Gender.
 - **Asymmetric**: one state rarer + more important. Disease test P/N. **Convention: rarest outcome = 1.**
- **Ordinal**: order meaningful, gap between successive values unknown. {small, medium, large}, grades, mineral hardness.
- **Numeric**: measurable quantity.
 - **Interval-scaled**: gaps meaningful, **no true zero**. Dates, temp in °C/°F.
 - **Ratio-scaled**: gaps and ratios meaningful, true zero. Temp in K, money, counts, age.

Older N-O-I-R split (Tan; DBK's slide)

- Nominal, Ordinal, Interval, Ratio. Binary folded into nominal.
- **Same content, different grouping**. Either scores; H&K's is safer as it names binary explicitly, which the similarity questions depend on.

Attribute Level	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function.	An attribute encompassing the notion of good, better, best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}.
Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

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2nd cut: discrete vs continuous

- **Discrete:** finite / countably infinite set. Zip codes, counts, words in a doc. Usually int.
 - **Binary** = special case of discrete.
 - **Symmetric binary:** both states equally valuable (gender).
 - **Asymmetric binary:** one state rarer + more important (disease test P/N). **Matters: Jaccard ignores 0-0 matches, SMC does not.**
- **Continuous:** real values. Temp, height, weight. Stored as float.

Types of datasets

- **Record**
 - *Data matrix:* m objects $\times$ n numeric attrs $\rightarrow$ points in n -dim space.
 - *Document data:* each doc = term vector, value = term frequency.
 - *Transaction data:* each record = set of items. Eg grocery basket.
- **Graph:** nodes + links. Eg WWW, molecular structures.
- **Ordered:** spatial (geo coords), temporal ($d = f(t)$), sequential, sequence data.

Characteristics of structured data

- **Dimensionality** $\rightarrow$ curse of dimensionality (see 2.2).
- **Sparsity:** % of cells not populated. Sparsity + density = 100.
- **Resolution:** patterns depend on the scale you look at.

2.2 Data Pre-processing

What is Data Pre-Processing? Explain the major tasks performed in it. 2+7 4+4 5+5

TOP 16/23 82 Bh, 81 Ba, 79 Ba, 75 Ch, 73 Ch, 72 Ka, 69 Ch Most-asked topic in the syllabus.

Why preprocess

- Real data is **dirty**: incomplete, noisy, inconsistent.
- Cause: huge size + multiple heterogeneous sources.
- **Garbage in $\rightarrow$ garbage out.** Low quality data $\rightarrow$ low quality mining.
- Data prep $\approx$ **60% of total effort** in KDD.

Measures of data quality 73 Shr asks this

- **Accuracy:** values correct.
- **Completeness:** no missing values.
- **Consistency:** no contradictions.
- **Timeliness:** updated in time.
- **Believability:** how far users trust it.
- **Interpretability:** how easily understood.

5 major tasks (this is the answer)

1. **Data cleaning:** fill missing, smooth noise, fix inconsistency, rm duplicates.
2. **Data integration:** merge multiple sources into 1 coherent store.
3. **Data transformation:** normalise, aggregate, generalise, construct attrs.
4. **Data reduction:** smaller volume, same analytical result.
5. **Data discretisation:** continuous $\rightarrow$ intervals (part of reduction).

Answer skeleton 7

1. Definition + "why": dirty data, garbage in garbage out.
2. Name the 5 tasks.
3. 2 lines each, with 1 concrete technique per task.
4. Close: cleaning + integration dominate the effort.

(a) Data cleaning

Missing values: why

- Equipment malfunction, misunderstanding at entry, deleted as inconsistent, not considered important, no change recorded.

How to handle missing data 74 Ash, 74 Ch

1. **Ignore the tuple:** usual when class label missing. Bad if % missing varies a lot.
2. **Fill manually:** tedious, infeasible at scale.
3. **Global constant:** eg "Unknown". Risk: mining may treat "Unknown" as a real class.
4. **Attribute mean** of all samples.
5. **Attribute mean of same class** → better than plain mean.
6. **Most probable value:** regression, Bayesian inference, decision tree. **Most popular + most accurate.**

Noisy data

- Noise = random error / variance in a measured variable.
- Cause: faulty instruments, entry error, transmission error, tech limits, naming inconsistency.

How to handle noise

- **Binning:** sort, partition into bins, smooth by bin *mean* / *median* / *boundary*.
- **Regression:** fit data to a function.
- **Clustering:** detect + rm outliers.
- **Combined computer + human:** machine flags suspicious, human checks.

Binning: 2 partitioning rules

- **Equal-width (distance):** N intervals of equal size.
- **Equal-depth (frequency):** N intervals, each with $\approx$ same no. of samples. Better for skewed data.

Equal-width bin width, A = lowest value, B = highest, N = no. of bins:

$$W = \frac{B - A}{N}$$

Your class notes print $W = (A - B)/N$, which gives a negative width. Wrong: it is $(B - A)/N$. verified/added

(b) Data integration 5

75 Ash

- Combines data from multiple sources into 1 coherent store.
- **Issues:**
 - **Entity identification:** is `cust_id` in A the same as `cust_no` in B? Solved w/ metadata.
 - **Redundancy:** an attr derivable from another. Detect by **correlation analysis** (χ^2 for nominal, correlation coefficient + covariance for numeric).
 - **Data value conflicts:** same entity, diff values across sources (diff units, scales, currency, representation).
 - **Duplication** at tuple level.
- Careful integration → reduces/avoids redundancy → improves mining speed + quality.

χ^2 test of independence (nominal pairs) NEW

- H_0 : the 2 attrs are **independent**. H_1 : associated (so 1 is redundant).
- Build the **contingency table** of observed counts O_{ij} .
- Expected count $E_{ij} = \frac{(\text{row}_i \text{ total}) \times (\text{col}_j \text{ total})}{\text{grand total}}$
- $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, $df = (r - 1)(c - 1)$.
- $\chi^2 >$ critical value at $\alpha \rightarrow$ **reject** $H_0 \rightarrow$ correlated $\rightarrow$ drop one attr.

Worked in one line

- Gender (M/F) vs preference (like/dislike), 100 people, all 4 margins = 50.
- Every $E_{ij} = \frac{50 \times 50}{100} = 25$. Observed 20/30/30/20.
- $\chi^2 = 4 \times \frac{5^2}{25} = 4$, $df = (2 - 1)(2 - 1) = 1$.
- Critical value at $\alpha = 0.05$, $df = 1$ is 3.841. $4 > 3.841 \rightarrow$ **reject** H_0 , gender and preference are associated.
- **Correlation $\neq$ causation.** Numeric analogue:
 $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$, formula in §2.4.

(c) Data transformation + normalisation 2+6 4 2+2+2

TOP /23 79 Ba, 78 Bh, 75 Ch, 73 Shr Normalisation numerical is a recurring 6-mark question → full solution in the Numerical companion.

Transformation methods

- **Smoothing:** rm noise.
- **Aggregation:** summarise, build data cube.
- **Generalisation:** climb concept hierarchy (street → city → country).
- **Normalisation:** scale into a small range.
- **Attribute construction:** build new attrs from given ones.

Why normalise

- Stops large-range attrs dominating small-range ones.
- Required by distance-based methods: KNN, K-means, ANN.

The 3 normalisation formulas (memorise exactly)

1. Min-max → maps to $[new_min, new_max]$:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

2. Z-score (zero-mean) → use when min/max unknown or outliers present:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \bar{A} = \text{mean}, \sigma_A = \text{std deviation}$$

3. Decimal scaling:

$$v' = \frac{v}{10^j} \quad j = \text{smallest int s.t. } \max(|v'|) < 1$$

Variant of 2: z-score w/ mean absolute deviation s_A , more robust to outliers than σ_A (deviations are not squared):

$$s_A = \frac{1}{n} \sum_i |v_i - \bar{A}| \quad v' = \frac{v - \bar{A}}{s_A}$$

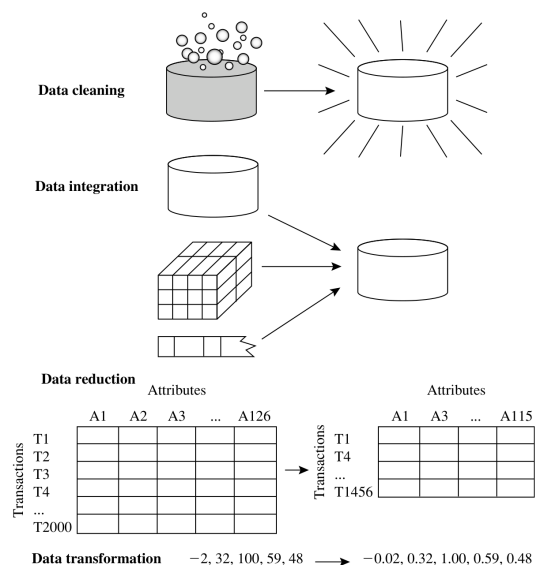
Error in your class notes verified/added : z-score is printed there as $v' = (v - min)/\sigma$. That is **wrong**. It is $(v - \text{mean})/\sigma$. Using min makes every $v' \geq 0$, which defeats the whole point of a zero-mean score.

(d) Data reduction 5

- Smaller volume, **same (or nearly same) analytical result**.
- **Strategies:**
 - **Sampling** (below)
 - **Dimensionality reduction:** PCA, SVD, wavelet
 - **Feature subset selection**
 - **Aggregation** / data cube
 - **Discretisation** + concept hierarchy
 - **Compression:** lossy / lossless
- **Han & Kamber's 3-way split** NEW (**safer skeleton for a 5-mark answer**):
 - **Dimensionality reduction:** wavelet transform, PCA, attribute subset selection. *Fewer attributes*.
 - **Numerosity reduction:** replace the data by a smaller representation. *Fewer tuples*.
 - *Parametric:* fit a model, store only its parameters → regression, log-linear models.
 - *Non-parametric:* store a reduced form directly → histograms, clustering, sampling, data cube aggregation.
 - **Data compression:** lossless (exact rebuild) vs lossy (approximation). The other two are themselves forms of compression.

Sampling types

- **Simple random:** equal prob for each item.
- **W/o replacement:** item removed once picked.
- **W/ replacement:** not removed, same item can repeat.
- **Stratified:** split into partitions, sample each. Best for skewed data.

The four forms at a glance

Cleaning, integration, reduction (attributes A1..A126 to A1..A115, tuples T1..T2000 to fewer), transformation ($-2, 32, 100 \rightarrow -0.02, 0.32, 1.00$). The one-figure answer to "forms of preprocessing". H&K Fig 3.1.

Feature subset selection

- Rm **redundant** features (purchase price + sales tax paid).
- Rm **irrelevant** features (student ID for predicting GPA).
- **Techniques:**
 - *Brute force*: try all subsets.
 - *Embedded*: selection happens inside the algorithm.
 - *Filter*: select before the algorithm runs.
 - *Wrapper*: use algorithm as a black box to score subsets.

Discretisation (feeds Ch 3 decision trees)

- Continuous → small no. of interval labels. Many algorithms need discrete input.
- **Unsupervised** (no class labels):
 - Equal-width, equal-frequency, **k-means** (cluster values, 1 bin per cluster).
- **Supervised** (uses class labels): **entropy-based**.
 - $H(p_1, \dots, p_s) = \sum_i p_i \log_2(1/p_i) = - \sum_i p_i \log_2(p_i)$
same quantity as *Info(D)* in Ch 3, §3.2 – written both ways in the literature
 - Split S at boundary T :
 $E(S, T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$
 - Pick T **minimising** E ; recurse until
 $Ent(S) - E(T, S) \leq \delta$.
- Also: **mapping to a new space** (Fourier, wavelet transform).

(e) PCA for dimensionality reduction 10 4

71 Shr 10, 76 Ch 4 numerical Numerical worked in the companion.

Key idea

- Find a **projection capturing the largest variance** in the data.
- New axes = eigenvectors of the **covariance matrix**, ordered by eigenvalue.
- Eigenvalue λ_i = variance carried by principal component i .
- Keep the top r components → drop the rest w/ little info loss.
- Components are **orthogonal** (uncorrelated).

Proportion of variance explained

$$\frac{\sum_{i=1}^r \lambda_i}{\sum_{i=1}^m \lambda_i} = \frac{\lambda_1 + \dots + \lambda_r}{\lambda_1 + \dots + \lambda_m}$$

- Denominator = trace of covariance matrix = total variance.

Curse of dimensionality 70 Ch 5, 81 Bh 6

- **Definition**: problems that appear only when no. of attrs (dims) is large; analysis that works in low dims fails in high dims.
- **Why**: volume of the space grows **exponentially** w/ dims.
 - Eg 10 bins per attr: 1-D = 10 cells, 2-D = 100, 10-D = 10^{10} cells.
 - Fixed no. of records → almost every cell empty → data **sparse**.
 - Records needed to cover the space grow exponentially too.
- **Distance concentration**:
 - $(d_{max} - d_{min})/d_{min} \rightarrow 0$ as dims $\uparrow$.
 - All points look **equidistant** → nearest neighbour meaningless.
- **Effects on DM**:
 - Clustering, KNN, outlier detection break (rely on density + distance).
 - Classifiers **overfit**: many attrs, few records per region.
 - Irrelevant + redundant attrs add noise.
 - Computation + storage cost $\uparrow$; data cannot be visualised.
- **Avoid by**:
 - dimensionality reduction: PCA, SVD, wavelet transform
 - feature subset selection: drop irrelevant/redundant attrs
 - feature construction: combine attrs into fewer, better ones
 - domain knowledge; subspace clustering

PCA process: 5 steps (DBK's sequence)

1. **Subtract the mean** from each dimension → zero-mean data.
2. **Compute the covariance matrix**.
3. **Compute eigenvalues + eigenvectors** of that matrix.
4. **Choose components, form feature vector**: sort λ descending, keep top r , put their eigenvectors in columns.
5. **Derive new data**: $\text{FinalData} = \text{RowFeatureVector} \times \text{RowZeroMeanData}$.

Notes

- Eigenvectors exist only for **square** matrices.
- An $m \times m$ matrix gives m of them.
- Eigenvectors of a **symmetric** matrix are mutually perpendicular.
- Normalise eigenvectors to unit length.
- Highly correlated dims → few large λ → PCA helps a lot.
 Uncorrelated dims → $r \approx m$ → **PCA does not help**.

Error in your class notes verified/added : the PCA section there lists “construct a neighbourhood graph” and “compute shortest-path / geodesic distances”. Those steps belong to **Isomap**, a non-linear method, **not** PCA. PCA is purely linear: covariance → eigenvectors.

2.3 OLAP & Multidimensional Data Analysis

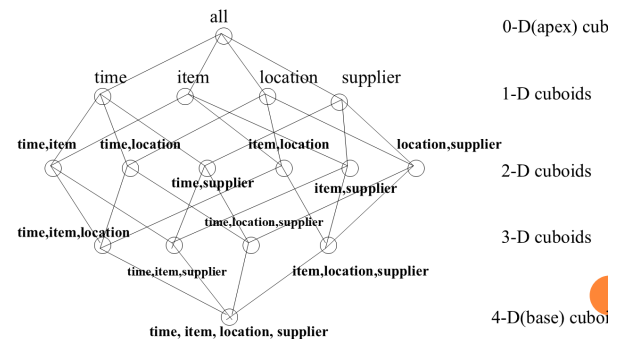
Explain typical OLAP operations over a multidim DW. Differentiate OLAP vs OLTP. 6+4 10 5+5 2+5+3

TOP 13/23 81 Bh, 79 Bh, 75 Ash, 74 Ch, 73 Ch, 72 Ch, 70 Asa

Multidimensional model

- **Dimension**: perspective the org keeps records by. Eg time, item, location, branch.
- **Measure**: the numeric quantity analysed. Eg units_sold, dollars_sold.
- **Fact table**: stores measures + FKs to dimension tables.
- **Data cube**: lets you view measures against a set of dims.
- **Cuboid**: the cube at one level of summarisation.
 - **Base cuboid** = lowest level, all n dims.
 - **Apex cuboid** = 0-D, highest summary, denoted all.
 - Lattice of cuboids = the full cube. n dims → 2^n cuboids.
- Schemas: **star**, **snowflake**, **fact constellation** → defined in Ch 1.
- **Concept hierarchy**: the levels of abstraction inside 1 dim, eg street < city < province < country.
 - This is **what roll-up and drill-down actually climb**.
 - Supplied manually by users/experts, or generated automatically from the data distribution.

Cube: A Lattice of Cuboids



3 kinds of measure, by aggregate function

- **Distributive**: partition the data, aggregate each part, aggregate the results → same answer. `sum()`, `count()`, `min()`, `max()`.
- **Algebraic**: computable from a bounded no. of distributive functions. `avg()` = `sum()/count()`, `min_N()`, `standard_deviation()`.
- **Holistic**: no constant-size sub-aggregate suffices. `median()`, `mode()`, `rank()`. **Expensive** → usually estimated.

The 5 OLAP operations (the core answer)

- **Roll-up (drill-up)**: summarise by climbing a hierarchy or reducing a dim. Eg city → country; or drop *time* entirely.
- **Drill-down (roll-down)**: reverse of roll-up. Summary → detail, or introduce a new dim. Eg quarter → month.
- **Slice**: select on **one** dim → sub-cube. Eg *time* = Q1.
- **Dice**: select on **two or more** dims. Eg *location* in {Kathmandu, Pokhara} **and** *time* in {Q1, Q2}.
- **Pivot (rotate)**: rotate axes to give an alternative presentation. 3-D → series of 2-D planes.
- **Others**: **drill-across** (across ≥ 2 fact tables), **drill-through** (down to the back-end relational tables).

OLAP vs OLTP (asked 4 times)**OLTP vs. OLAP**

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date, detailed, relational, isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

OLAP server types (detail in Ch 1, DW architecture)

- **ROLAP**: relational back end. Scales best.
- **MOLAP**: multidim array store. Fastest, wastes space on sparse data.
- **HOLAP**: hybrid. Detail relational + aggregates multidim.

Base & apex cuboid, slice & dice, roll-up & drill-down w/ example 2+3+3+3

76 Ash Answer = the definitions above + 1 worked example each. Cuboid-count and schema-drawing questions are worked in the Numerical companion
78 Bh, 76 Ch.

2.4 Various Similarity Measures**How is similarity/dissimilarity calculated? Properties of a distance metric. 3 10 4**

HOT 7/23 80 Bh, 80 Ba, 78 Bh, 76 Ch, 76 Ash, 71 Ch, 70 Asa **Almost always numerical** → companion.

Similarity vs dissimilarity

- **Similarity**: how alike 2 objects are. *Higher* = more alike. Usually $[0, 1]$.
- **Dissimilarity / distance**: how different. *Lower* = more alike. Min = 0, upper limit varies, often $[0, \infty]$.
- Used for: clustering, KNN, outlier detection, search, recommendation, near-duplicate detection.

Distance metric: 4 properties (71 Ch asks exactly this, 10)

- 1. Non-negativity**: $d(x, y) \geq 0$.
- 2. Identity**: $d(x, y) = 0$ iff $x = y$.
- 3. Symmetry**: $d(x, y) = d(y, x)$.
- 4. Triangle inequality**: $d(x, y) \leq d(x, z) + d(z, y)$.
 - Triangle inequality → guarantees the function is *well behaved*: the direct connection is the shortest.
 - A measure satisfying all 4 = a **metric**. Minkowski distances are metrics for $r \geq 1$.
 - **Subtle point worth 1 mark**: non-negativity is *implied* by the other 3, so strictly only 3 are independent.
 - **Weighted Euclidean** (attrs of unequal importance):

$$d(i, j) = \sqrt{\sum_k w_k (x_{ik} - x_{jk})^2}.$$

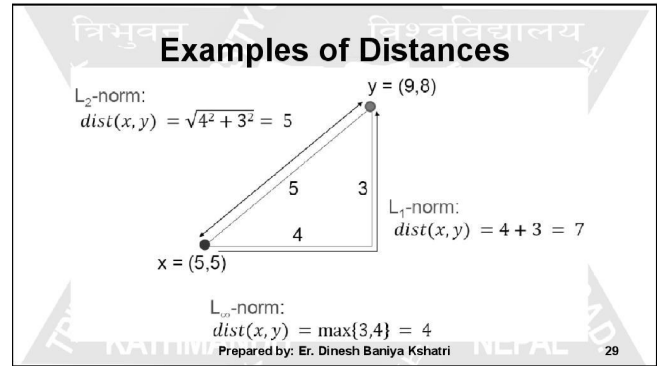
Minkowski family: the one formula that generates the rest

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- n = no. of dims, r = parameter, $r \geq 1$. **Do not confuse r with n .**
- **Notation clash:** H&K call the parameter h and reserve p for the no. of attrs; other books write L_p where their p is this r . **Define your symbols.**
- $r = 1 \rightarrow$ **Manhattan** (L_1 , city block, taxicab). Binary case = **Hamming distance**.
- $r = 2 \rightarrow$ **Euclidean** (L_2).
- $r \rightarrow \infty \rightarrow$ **Supremum** ($L_{\max}$, L_∞ , Chebyshev, chessboard) = max difference in any single coordinate.

Written out

- Euclidean: $\sqrt{\sum_k (p_k - q_k)^2}$
- Manhattan: $\sum_k |p_k - q_k|$
- Supremum: $\max_k |p_k - q_k|$



Minkowski Distance – Example

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

I.1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

I.2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

I.3	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

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Binary attributes: SMC vs Jaccard 2+2+2

Build a 2×2 contingency count for objects i, j . **Two notations exist, know both:**

- $q = M_{11}$ = attrs where both = 1 $t = M_{00}$ = attrs where both = 0
- $r = M_{10} = i=1, j=0$ $s = M_{01} = i=0, j=1$
- Total $p = q + r + s + t$. **H&K use q, r, s, t ; Tan and DBK use M_{11} etc.**

$$d = \frac{r + s}{q + r + s + t} \quad d = \frac{r + s}{q + r + s} \quad SMC = \frac{q + t}{q + r + s + t} = 1 - d_{\text{sym}}$$

symmetric asymmetric = 1 - d_{sym}

- **Only difference: Jaccard/asymmetric drop t (the 0-0 matches).**
- **SMC** $\rightarrow$ **symmetric** binary (a 0-0 match is informative).
- **Jaccard** $\rightarrow$ **asymmetric** binary (0-0 meaningless: 2 people both lacking a rare disease says nothing).
- Sets form: $J(A, B) = |A \cap B| / |A \cup B|$.
- **Read the verb.** "Find the Jaccard coefficient" wants J ; "find the dissimilarity" wants $d = 1 - J$.

SMC vs. Jaccard Example

$p = 1000000000$
 $q = 0000001001$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)
 $M_{10} = 1$ (the number of attributes where p was 1 and q was 0)
 $M_{00} = 7$ (the number of attributes where p was 0 and q was 0)
 $M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0 + 7) / (2 + 1 + 0 + 7) = 0.7$

$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$

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Nominal attributes

$$d(i, j) = \frac{p - m}{p} \quad \text{sim}(i, j) = 1 - d(i, j) = \frac{m}{p}$$

- m = no. of attrs where i and j are in the **same state**, p = total attrs.
- Alternative: encode each of the M states as an **asymmetric binary** attr, then use Jaccard.

Ordinal attributes

1. Replace value x_{if} by its **rank** $r_{if} \in \{1, \dots, M_f\}$.
2. Normalise rank to $[0, 1]$ so every attr has equal weight:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

3. Treat z_{if} as **numeric** → use Euclidean / Manhattan.
- Eg {fair, good, excellent}, $M_f = 3 \rightarrow$ ranks 1,2,3 → $z = 0, 0.5, 1$.

Cosine similarity 3

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}$$

- $d_1 \cdot d_2$ = dot product, $\|d\| = \sqrt{\sum d_k^2}$ = vector length.
- Range $[0, 1]$ for non-negative vectors.
- Aligned → angle $0^\circ \rightarrow \cos = 1$. Orthogonal (no common terms) → $90^\circ \rightarrow \cos = 0$.
- **Ignores magnitude, measures direction only** → ideal for documents, where doc length shouldn't matter.

Which measure for which data (picking wrong = losing the marks)

Data	Use
Numeric, scale matters	Euclidean / Manhattan / Minkowski
Numeric, only direction matters (documents)	Cosine
Symmetric binary	SMC
Asymmetric binary, sets	Jaccard
Nominal / categorical	Simple mismatch $d = (p - m)/p$, m = no. of matches, p = total attrs
Strings	Hamming (equal length) / Levenshtein edit distance
Testing linear relationship	Correlation coefficient

Data matrix vs dissimilarity matrix

- **Data matrix** (n objects $\times p$ attrs): stores the objects themselves. **Two-mode** (rows = objects, cols = attrs).
- **Dissimilarity matrix** ($n \times n$): stores $d(i, j)$ for every pair. **One-mode** (both axes are objects).
- Properties: $d(i, i) = 0$, $d(i, j) = d(j, i) \rightarrow$ only the **lower triangle** is written.
- Most clustering + KNN algorithms run on the dissimilarity matrix.

Mixed attribute types

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- $\delta_{ij}^{(f)} = 0$ if the attr is missing for either object, or it is asymmetric binary and both are 0. Else $\delta = 1$.
- $d_{ij}^{(f)}$ computed per attr type, each scaled to $[0, 1]$ (numeric: divide by the range).

Correlation coefficient

$$\text{corr}(x, y) = \frac{\sum_i (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum_i (x_i - \mu_x)^2} \sqrt{\sum_i (y_i - \mu_y)^2}} = \frac{s_{xy}}{s_x s_y}$$

- = cosine similarity of the **mean-centred** vectors.
- Range $[-1, 1]$: +1 positive, -1 negative, 0 no correlation.
- **Covariance** $s_{xy} = \frac{1}{n-1} \sum_k (x_k - \bar{x})(y_k - \bar{y})$.
- Correlation = scaled (unit-free) version of covariance.

2.5 Basic Statistical Descriptions of Data

NEW Han & Kamber 3rd ed §2.2. **Never asked in the 23 papers**, so no tier chip. Added because PC's 2025–26 Unit 2 handout and its numerical companion are built entirely on it, and because the IQR outlier rule below is the statistical basis of the boxplot test used in Ch 6. **Learn the two boxed formulas; skim the rest.**

Measures of central tendency

- **Mean** $\bar{x} = \frac{1}{n} \sum_i x_i$ (sample), $\mu = \frac{1}{N} \sum x$ (population).
- **Weighted mean** $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$.
- **Trimmed mean**: chop the extreme values first → kills outlier sensitivity. Chop at most 2% per end.
- **Median**: middle value if n odd, mean of the middle two if n even.
- **Median is holistic** → needs a full sort → expensive on a large table. (Same holistic/distributive/algebraic split as the cube measures, §2.3.)
- **Mode**: most frequent value. Unimodal / bimodal / trimodal / multimodal. No mode if every value occurs once.
- **Midrange** = $\frac{\max + \min}{2}$. Cheap: 1 pass, $\max()$ and $\min()$.

Median of grouped data (intervals, not raw values):

$$\text{median} = L_1 + \left(\frac{N/2 - (\sum \text{freq})_l}{\text{freq}_{\text{median}}} \right) \times \text{width}$$

L_1 = lower boundary of the median interval, N = total count, $(\sum \text{freq})_l$ = cumulative frequency of all intervals *below* it, $\text{freq}_{\text{median}}$ = its own frequency, width = its width.

Locate the interval first: the one where cumulative frequency first reaches $N/2$.

Empirical relation, unimodal and moderately skewed:

$$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$$

$$\Rightarrow \text{mode} \approx 3\text{median} - 2\text{mean}$$

Skew, read off mean vs median vs mode

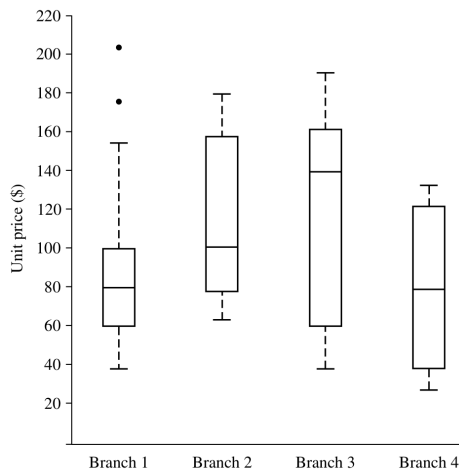
- **Symmetric**: $\text{mean} = \text{median} = \text{mode}$.
- **Positively skewed** (right tail): $\text{mode} < \text{median} < \text{mean}$.
- **Negatively skewed** (left tail): $\text{mean} < \text{median} < \text{mode}$.
- **Mean is dragged toward the tail**, median is not. That is why median is the robust one.

Measures of dispersion

- **Range** = $\max - \min$.
- **Quartiles**: Q_1 = 25th percentile, Q_2 = median, Q_3 = 75th percentile.
- **IQR** = $Q_3 - Q_1$ = spread of the middle half.
- **Variance** $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$, **std deviation** $s = \sqrt{s^2}$.
- Population forms use $\frac{1}{N}$ and σ .

Two quartile conventions (they disagree, say which you used)

- **Positional (Han & Kamber, the syllabus book):** for $n = 12$ take the 3rd, 6th and 9th sorted values directly.
- **Interpolated (PC's class notes):** position of $Q_k = \frac{k(n+1)}{4}$, then linear interpolation between the two neighbouring values.
- Both are marked correct. **The outlier verdict comes out the same either way** → see Ch 2 Numerical, M7.



Four boxplots side by side: box Q_1 to Q_3 , median line, whiskers, and Branch 1's two outliers plotted as lone points above its whisker. H&K Fig 2.3.

Five-number summary and boxplot

- **Five-number summary**, in this order:
Minimum, Q_1 , Median, Q_3 , Maximum.
- **Boxplot:** box spans Q_1 to Q_3 (box length = IQR), line inside marks the median, two **whiskers** extend to min and max, outliers plotted as individual points.

IQR outlier rule (the one to memorise): a value is a suspected outlier if

$$x < Q_1 - 1.5 IQR \quad \text{or} \quad x > Q_3 + 1.5 IQR$$

Beyond $3 IQR$ = **extreme outlier**. **Used again in Ch 6** as the boxplot test.

Graphic displays

- **Histogram:** x = value / bin, y = frequency.
- **Quantile plot:** each x_i against f_i , the fraction of data at or below it.
- **Quantile-quantile (q-q) plot:** quantiles of one distribution against the corresponding quantiles of another → compares 2 datasets.
- **Scatter plot:** each pair as a point → shows correlation and clusters.
- **Histogram often tells more than a boxplot:** two datasets can share a boxplot and have completely different histograms.

2.6 PYQ Mapping

Tiers: **TOP /23** = 9+/23 **HOT /23** = 5–8 **PIN /23** = 1–4 but ≥ 8 marks. **N** marks a question fully solved in the Ch 2 Numerical companion.

2.1 Data Types and Attributes

1. **PIN 1/23** "What is data mining? Explain different data types of attributes in a dataset." 10 71 Shr

2.2 Data Pre-processing

2. **TOP 12/23** **Why preprocess + major tasks / methods.**
 - "What is Data Pre-Processing? Briefly explain the major tasks performed." 2+7 81 Ba
 - "Why data preprocessing is necessary? Explain methods to maintain data quality." 4+4 75 Ch
 - "Why is it required? Explain approaches of data cleaning." 5+5 72 Ka
3. **PIN 1/23** "Measuring elements of data Quality? Data transformation by normalization w/ example." 2+6 73 Shr
4. **N** "Use the following methods to normalize 200, 300, 400, 600, 1000" (min-max, z-score, decimal scaling). 2+2+2 78 Bh
5. **PIN 2/23** Handling missing values. 5 74 Ash 2 74 Ch
6. **PIN 1/23** Issues in Data Integration. 5 75 Ash
7. **PIN 1/23** "What is 'curve of Dimensionality'? How can it be avoided?" 5 70 Ch paper typo: "curse"
8. **PIN 1/23** "Discuss the impact of noisy data in data mining." 5 70 Ch
9. **PIN 1/23** "How can PCA be used for dimensionality reduction?" 10 71 Shr
10. **N** "Find the principal components + proportion of total variance" (3×3 covariance matrix). 4 76 Ch
11. **PIN 3/23** Short notes: data smoothing 76 Ash · data normalization 75 Ch · data transformation 73 Shr, 72 Ka. 3 4

2.3 OLAP & Multidimensional Data Analysis

12. **TOP 10/23** **OLAP operations + OLAP vs OLTP.**
 - "Explain typical OLAP operations...Differentiate OLAP and OLTP tools." 6+4 79 Bh
 - "How do you perform analysis of multidimensional data? Explain w/ OLAP." 10 72 Ch
 - "Discuss Data Integration issues. Describe OLAP + operations w/ example." 5+5 75 Ash
 - "Approaches to handle missing data? OLAP + operations? OLAP vs OLTP." 2+5+3 74 Ch
13. **PIN 1/23** "Dimensional data? Base & apex cuboid? Slicing & Dicing? Roll Down/Up? Give example." 2+3+3+3 76 Ash

14. **N** Star schema for a 4-dim DW + which OLAP ops list total charge. 3+3 78 Bh
15. **N** Snowflake schema for a 5-dim DW + OLAP ops for count per state per year. 3+3 76 Ch
16. **PIN 3/23** Short notes: OLAP Operations 74 Ash, 73 Shr · OLAP and OLTP 75 Ch. 4

2.4 Various Similarity Measures

17. **N** Cosine sim (Obj 2,4) + Euclidean (1,3) and (1,4), 4-attribute object table. 3+3+3 80 Bh
18. **N** Jaccard (asymmetric) + symmetric-binary dissimilarity + SMC + cosine of 2 doc vectors. 2+2+2+1 80 Ba
19. **N** Manhattan + Supremum distance matrix · cosine of 2 vectors · Jaccard + SMC of 2 binary vectors. 2+1+2 76 Ch
20. **PIN 1/23** "Properties a Distance Metric needs to support w.r.t. Minkowski's distance." 10 71 Ch
21. **PIN 1/23** "List types of similarity measures + their application areas." 3 80 Ba
22. **PIN 2/23** Short notes: Minkowski Distance 78 Bh · similarity measures between tuples 76 Ash. 4 3

Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. **verified/added**

- What is performed during preprocessing? + techniques to handle missing values. 4+4 82 Bh
- **Curse of dimensionality** in data mining. 6 81 Bh
- Why are similarity measures important? + the different measures w/ examples. 2+6 81 Bh
- **N** Bin means at depth 3 + min-max + z-score + decimal scaling of 35, on 27 values. 6 79 Ba
- Nominal and ordinal attributes + handling missing and noisy data. 6 79 Ba
- Preprocessing + **data sampling** + dimensionality reduction w/ example. 2+4+4 73 Ch
- How does real data become **asymmetric**? Symmetric vs asymmetric data. 5 73 Ch
- What is dimensionality reduction and why does it matter? 5 70 Asa
- Properties of a distance metric (its second half is §3.4). 10 70 Asa
- Different data types w/ examples 69 Ch · short note **OLAP cubes** 70 Asa. 5 5

Coverage check

- All Ch 2 PYQs covered. 8 are numerical → fully solved in the companion.
- **3 errors in your source notes corrected here** **verified/added** :
 - Z-score written as $(v - \min)/\sigma \rightarrow$ must be $(v - \text{mean})/\sigma$.
 - Equal-width bin width $W = (A - B)/N \rightarrow$ must be $(B - A)/N$.
 - PCA section contaminated w/ Isomap steps (neighbourhood graph, geodesic distances).
- **Gap filled by research** **verified/added** : similarity-measure formulas were absent from the typed notes (they say only "Refer Class Note"), so they were taken from DBK Lecture-2 and then **verified against Han & Kamber 3rd ed** (the syllabus reference book).

Sources used for this chapter

- **Han & Kamber, 3rd ed** (full textbook PDF): ch 2 attribute types + all proximity measures, ch 3 preprocessing + normalisation, ch 4 data cube + OLAP. **Primary authority.**
- **DBK Lectures 1-4** (Thapathali class slides): attribute types, binning example, discretisation, distance/similarity, PCA process, data cube + concept hierarchies.
- data_mining_/Chapter 1_2.pdf (Bhupesh K. Mishra), Chapter3_0.ppt, Chapter 2_Data Warehouse.ppt.
- **Notes by PC** (Er. Prakash Chandra Prasad, Asst. Prof., IOE Pulchowk), dated 2025-26: Unit 2.docx, unit 2 Numerical.docx, Redundancy and Correlation Analysis.docx. Source of the new §2.5 and of the χ^2 procedure in §2.2(b). **Newest teaching material available**, so treat §2.5 as insurance even though no past paper has asked it.
- Where sources disagree, **Han & Kamber 3rd ed wins** and the clash is flagged inline.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.